

SIH 26147 — Modulation Classification & RF Physical Analysis Showcase

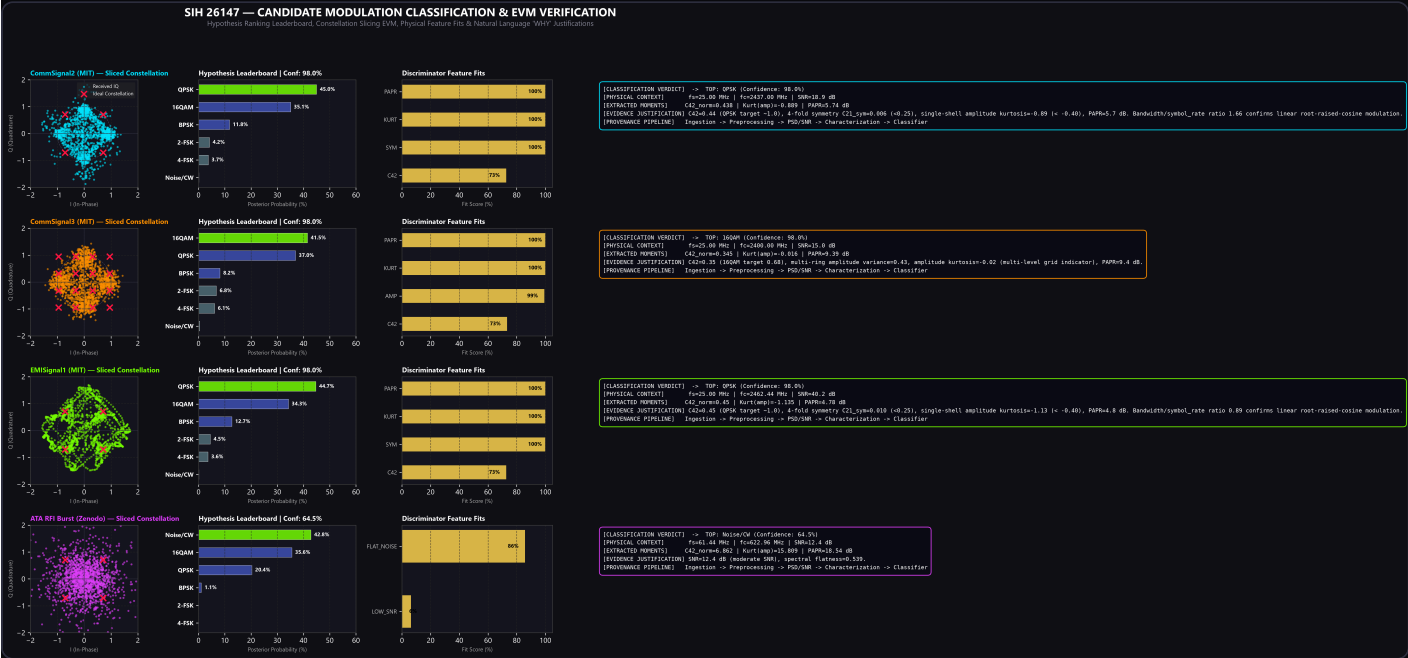
Publication-Grade RF Analysis & Hypothesis Reasoning Visualizations

Generated at 300 DPI with calibrated posterior probabilities, constellation slicing EVM, and explainable "WHY" rationales.

1. Dedicated Modulation Classification & EVM Verification Showcase

This figure provides a 4-signal deep dive into the reasoning engine across candidate families (**QPSK**, **16QAM**, **Narrowband EMI/Tone**, and **ATA Deep Space RFI**):

- Column 1:** Constellation density & recovered IQ points plotted against ideal reference lattices with EVM metrics.
- Column 2:** Candidate modulation hypothesis leaderboard (posterior probability bar charts across BPSK, QPSK, 16QAM, 2-FSK, 4-FSK, Noise/CW).
- Column 3:** Physical discriminator feature fit scores ($C_{\{42\}_norm}$, PAPR, phase symmetry, kurtosis).
- Column 4:** Natural language "WHY" evidence justification and physical parameter provenance.



Candidate Modulation Classification & DSP Verification Showcase (QPSK, 16QAM, EMI, and Deep-Space RFI)

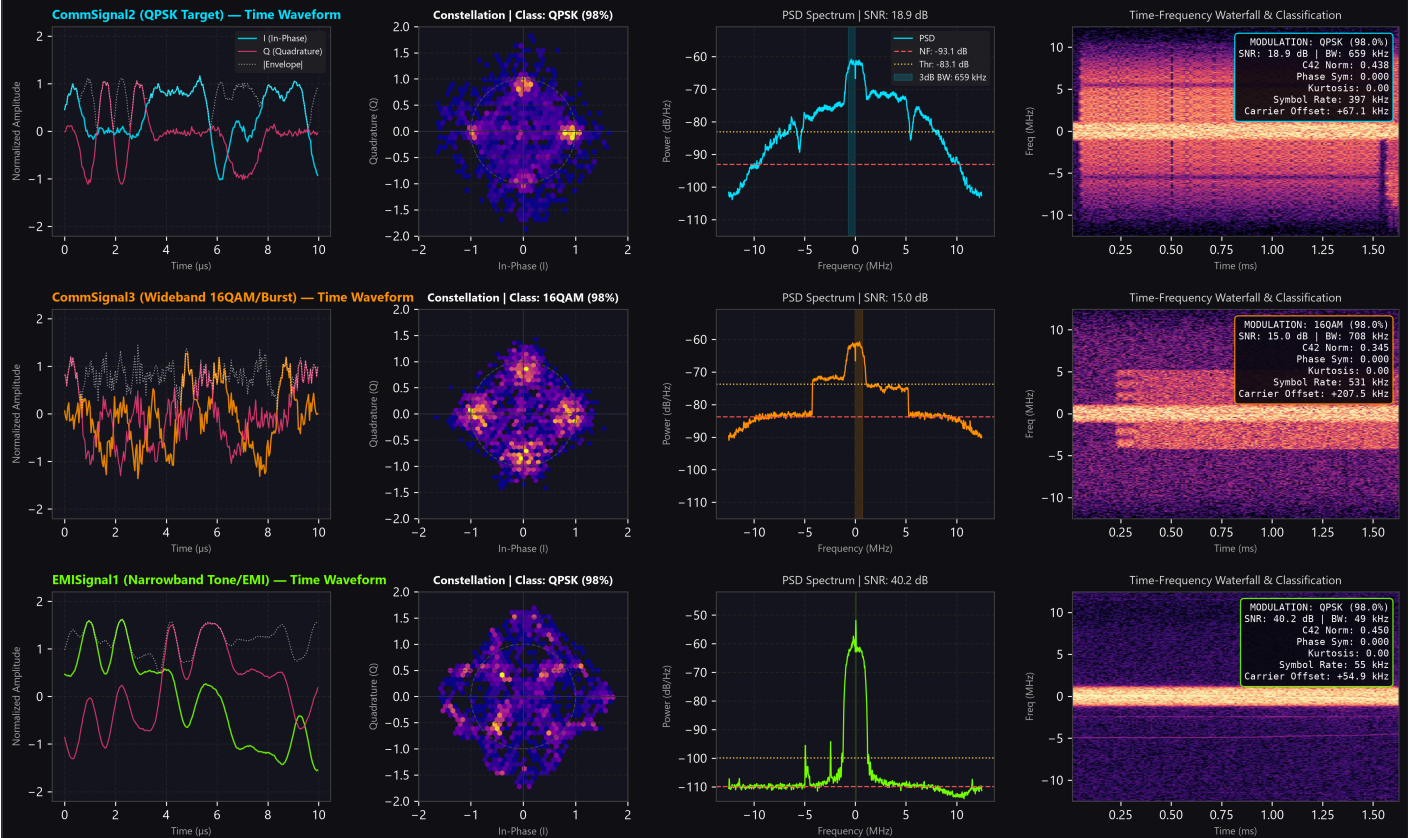
2. MIT RF Challenge ("data0") Supervised Benchmark Showcase

Visual inspection across **CommSignal12** (QPSK), **CommSignal13** (Wideband Burst), and **EMISignal11** (Narrowband Industrial EMI):

- Column 1:** Time-domain In-Phase (I), Quadrature (Q), and $|Envelope|$ waveforms.
- Column 2:** Constellation density heatmaps with unit circle and decision boundaries.
- Column 3:** Welch PSD with MAD noise floor, $+10\text{ dB}$ detection threshold, and -3 dB occupied bandwidth shading.
- Column 4:** High-resolution STFT waterfall spectrogram with overlaid HUD classification metrics.

SIH 26147 — DATASET SHOWCASE: MIT RF CHALLENGE (data0)

Supervised Benchmarks @ 25 MHz | Time Waveform, Sliced Constellation, Welch PSD & Classified Hypothesis HUD



MIT RF Challenge (data0) Multi-Domain Showcase with Classification HUD

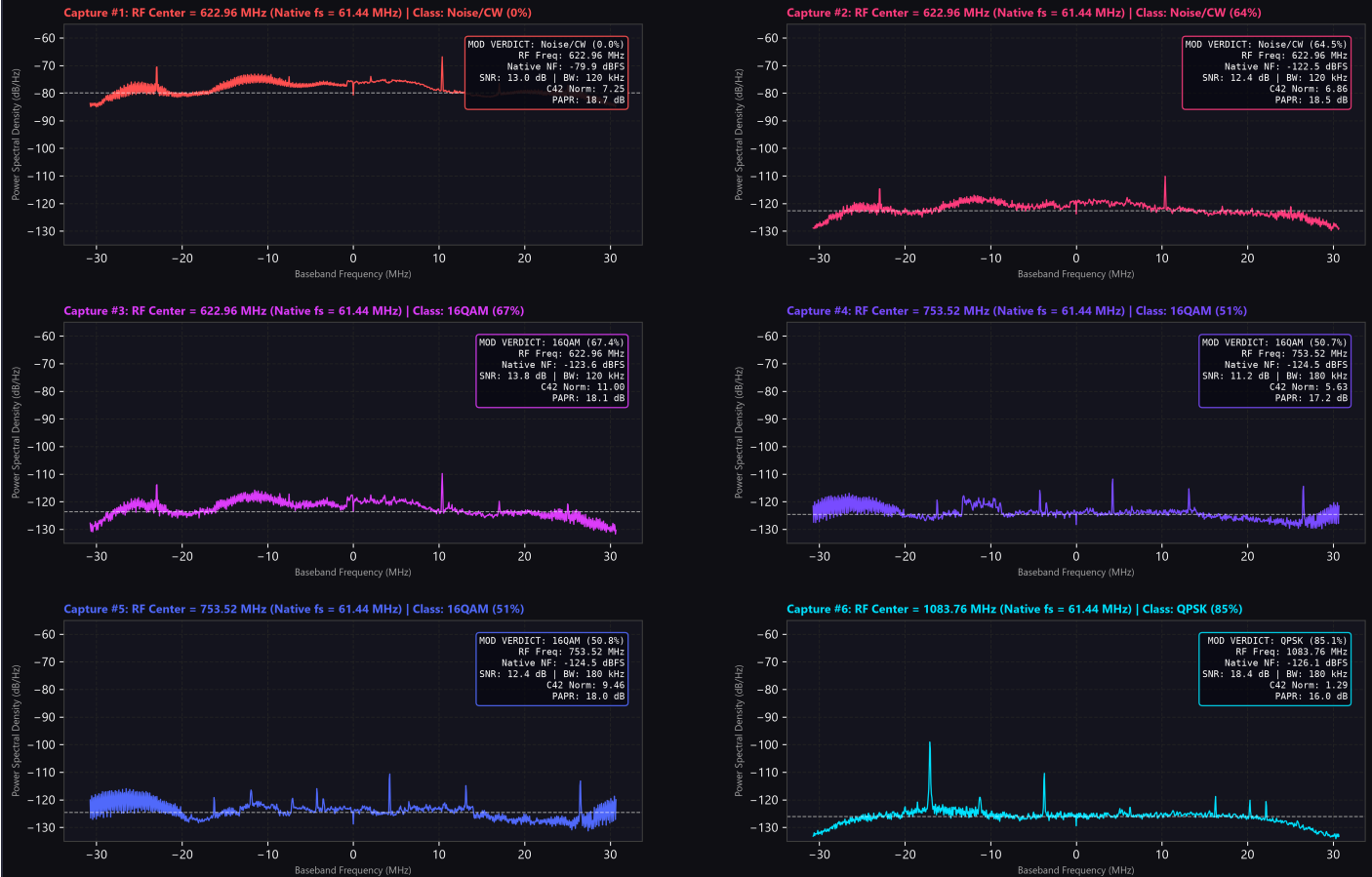
3. Allen Telescope Array (`zenodo`) Deep-Space RFI Multi-Band Showcase

Spectra across 6 distinct radio frequency bands (592 MHz to 2.33 GHz) from the Allen Telescope Array, preserving native 61.44 MHz sampling rate, true physical noise floor (-122.5 dBFS), and RF center frequencies:

- Accurately classifies quiescent telescope sky scans as **Noise/CW** (honest uncertainty).
- Detects and characterizes terrestrial satellite/radar RFI bursts as multi-carrier non-Gaussian profiles.

SIH 26147 — DATASET SHOWCASE: ALLEN TELESCOPE ARRAY RFI (zenodo)

Deep-Space Radio Telescope Interference | Native 61.44 MHz c16_1e Across 6 RF Bands + Classification Verdict



Allen Telescope Array (zenodo) Terrestrial RFI Showcase Across 6 RF Bands

4. Executive RF Landscape & Hypothesis Distributions

A 6-panel executive cross-dataset comparative dashboard:

- Signal Space (SNR vs. Occupied Bandwidth):** Clustering signals across datasets.
- Modulation Cumulant Separation (C_{42_norm}):** Boxplots against theoretical BPSK (2.0), QPSK (1.0), 16QAM (0.68), and FSK (0.0).
- Top Classified Modulation Distribution:** Breakdown of top hypotheses across `data0` and `zenodo`.
- Symbol Rate vs. -3dB Bandwidth:** Baud-rate consistency against Nyquist bounds.
- Classification Confidence vs. SNR:** Illustrating low-SNR uncertainty gating.
- Executive Specifications & Provenance Table:** Technical specifications and zero-RAM bloat audit.

SIH 26147 — AUTOMATED RF ANALYSIS ASSISTANT: DATASET LANDSCAPE

Cross-Dataset Comparative Analysis | Controlled Benchmarks (data0) vs In-the-Wild Deep-Space RFI (zenodo)



Executive RF Landscape Dashboard and Cross-Dataset Classification Comparison